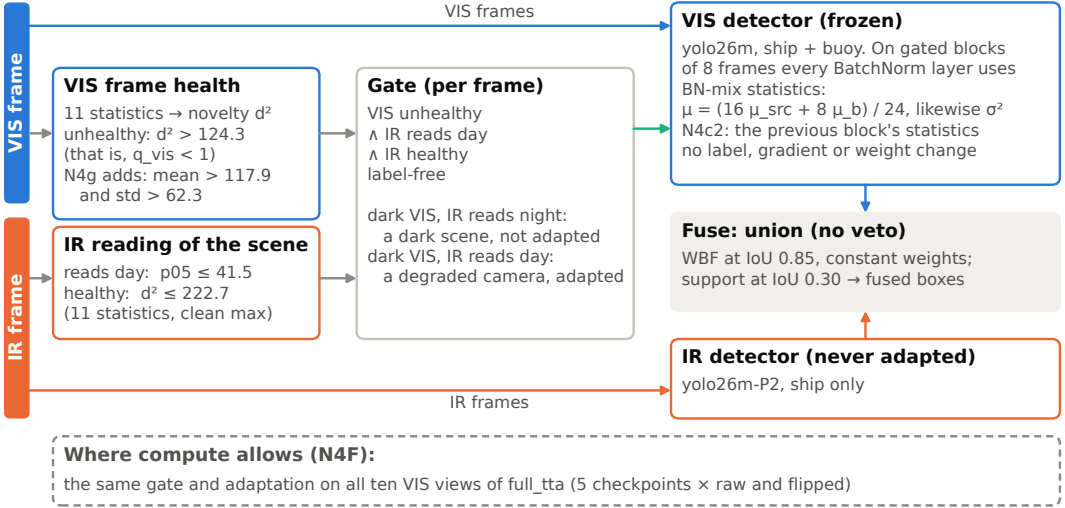
VIS frame
IR frame
VIS frames
VIS detector (frozen)
yolo26m, ship + buoy. On gated blocks
of 8 frames every BatchNorm layer uses
BN-mix statistics:
μ = (16 μ_src + 8 μ_b) / 24, likewise σ²
N4c2: the previous block's statistics
no label, gradient or weight change
VIS frame health
11 statistics → novelty d²
unhealthy: d² > 124.3
(that is, q_vis < 1)
N4g adds: mean > 117.9
and std > 62.3
Gate (per frame)
VIS unhealthy
∧ IR reads day
∧ IR healthy
label-free
dark VIS, IR reads night:
a dark scene, not adapted
dark VIS, IR reads day:
a degraded camera, adapted
IR reading of the scene
reads day: p05 ≤ 41.5
healthy: d² ≤ 222.7
(11 statistics, clean max)
Fuse: union (no veto)
WBF at IoU 0.85, constant weights;
support at IoU 0.30 → fused boxes
IR detector (never adapted)
yolo26m-P2, ship only
IR frames
Where compute allows (N4F):
the same gate and adaptation on all ten VIS views of full_tta (5 checkpoints × raw and flipped)